

The influence of channel geometry on the pressure drop in automotive catalytic converters: Model development and validation

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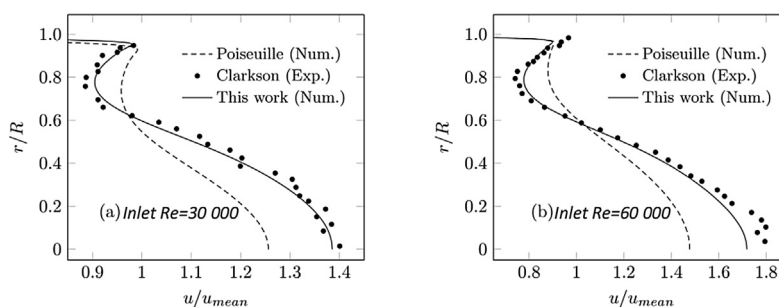
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HIGHLIGHTS

- A fully predictive multiscale pressure drop model for monolith substrates is presented.
- Excellent agreement with experimental velocity profiles was observed.
- The model works for several channel shapes and substrate void fractions.
- The methodology has general applicability to substrates with other channel shapes.

GRAPHICAL ABSTRACT



ARTICLE INFO

Article history:

Received 7 May 2019

Received in revised form 19 September 2019

Accepted 23 October 2019

Available online 4 November 2019

Keywords:

CFD
Monolith
Permeability
Pressure drop
Catalytic converter
Channel shape

ABSTRACT

This paper presents a detailed physics based model for the pressure drop through a honeycomb substrate for several channel shapes and void fractions. A CFD-based computational model of a single channel is used to study the pressure drop when flow is entering, passing through and leaving the substrate. An extensive set of 3D computational experiments covering square, hexagonal and triangular channel cross-sections, void fractions from 0.39 to 0.84 and channel Re from 95 to 1284 is used. It is shown that altering the void fraction changes the pressure drop at the inlet and outlet of the substrate, however, its effect on the friction factor inside the substrate is minor. The resulting model can be used either as a semi-empirical lumped model for pressure drop and in 3D full-scale simulations with a porous medium representing the substrate. A validation for the velocity profile in a full scale monolith with experimental data available in the literature is carried out and an excellent agreement is observed. The proposed model significantly improves the prediction of the flow distribution across the substrate, which has remained unaddressed historically by existing models and is the most important effect required to make accurate predictions of heat distribution, conversion efficiency and others in full-scale simulations.

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1. Introduction

Monolith reactors are extensively used in several industrial systems such as NO_x reduction, H₂ production, CH₄ autothermal reforming and especially as the core part of automotive catalytic converters (Chen et al., 2008; Dai et al., 2013; Quintanilla et al.,

2018; João Regufe et al., 2019; Thakkar et al., 2018; Roy et al., 2019). Typically, monoliths have a honeycomb type structures with many parallel channels. The catalyst is usually contained in a washcoat applied to the surface of the channels. A schematic of a monolithic catalytic converter is shown in Fig. 1.

Modelling a catalytic converter involves many length scales, all related at some level. The smallest scale is the molecular level, where the chemical reactions occur. It is followed by the diffusion inside the washcoat pores that contains the catalysts. At a channel

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Nomenclature

c_1, n	permeability model constants	Re	channel Re number
C_u	inertial resistance factor	u_i	velocity vector, m/s
D_H	channel hydraulic diameter, m	u_c	channel velocity, m/s
f	friction coefficient	x^+	dimensionless axial coordinate
Gz_i	inverse Graetz number	α_i	substrate apparent permeability, $1/m^2$
L	channel characteristic length, m	α_{afp}	apparent face permeability, m
L_H	channel hydraulic entrance length, m	δ_{ij}	Kronecker delta
L_s	substrate length, m	δm	apparent porous jump thickness, m
L_w	substrate wall thickness, m	ϕ	monolith void fraction
P	pressure, Pa	ρ	density, kg/m^3
P_s	static pressure, Pa	μ	viscosity, $kg/m\cdot s$
P_d	dynamic pressure, Pa	fd	sub-index standing for fully developed flow
Pr	Prandtl number		

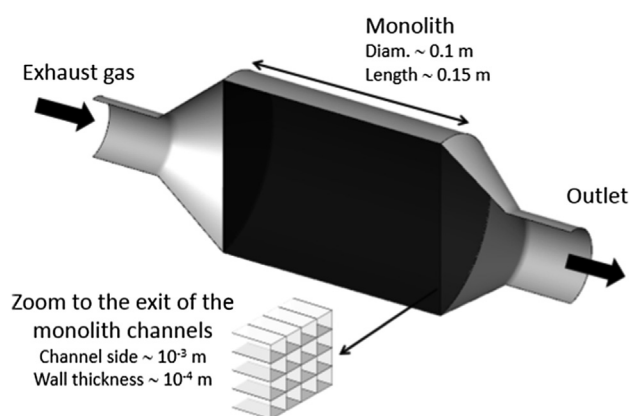


Fig. 1. Schematic of a catalytic converter. Source Cornejo et al. (2018c). Reprinted with permission of Elsevier

scale, the transport of species from the bulk to the surface of the washcoat is defined. The inlet conditions for the channels are given at a converter scale, also many important variables, such as the heat losses, ageing and pressure drop depend on the flow distribution across the substrate (Benjamin et al., 1996; Kim et al., 2009; Kim and Kim, 2007). For this reason, reliable models of the whole reactor are necessary for design improvement and process optimisation (Grimm and Mazumder, 2008; Kostoglou et al., 2011). At a full scale, honeycomb substrates are modelled as a continuum, because accounting for all the channels individually is computationally prohibitive (Bertrand et al., 2012; Kumar and Mazumder, 2010). That approach emphasizes the macro scale fluid dynamics at the expense of details at a washcoat and channel level. There are several differences between an actual honeycomb and a homogeneous porous medium, hence models that are physically consistent at a multiscale level are an extensive topic of research. The cornerstone for a reliable converter models is the apparent permeability of the porous medium representing the monolith. The simplest permeability model results when Poiseuille flow is assumed. By matching the Hagen-Poiseuille equation to Darcy's law, a single and constant apparent permeability can be obtained. It neglects any contraction and expansion of the flow at the beginning and end of the substrate, and the flow is assumed fully developed from the very inlet of the channels. The model is easy to implement, however, it fails in predicting the flow distribution and total pressure drop accurately (John Clarkson, 1995). An apparent permeability model that accounts for the effect of the upstream

turbulence was proposed by Ekstrom and Andersson (2002). In that model an average apparent permeability was obtained by curve fitting a series of experiments of pressure drop through a converter. Separately, Benjamin et al. (2001) modelled the extra pressure drop due to the flow entering at an angle to the substrate. According to their results, accounting for that extra resistance significantly affects on the prediction of the flow distribution significantly. Regarding the hydraulic entrance region, the correlation for the friction factor of developing flow in pipes proposed by Shah (1978) applied to a honeycomb was also tested, finding that it is not sufficient to predict the flow mal-distribution inside a converter (Benjamin et al., 1996). It should be noted that the velocity profile at the inlet of the honeycomb channel is not flat due to the reduction in the flow area at the frontal face of the substrate. Hence, models for developing flow in pipes starting from a flat profile may not apply directly (Lobo and Chatterjee, 2018). In this work, the model proposed by Cornejo et al. (2019b) was adopted and expanded to several substrate channel shapes. This model was preferred because it accounts for the losses when the flow is entering, developing, developed and exiting the substrate, which is especially important for modelling and control systems using several substrates (Wendland and Matthes, 1986; Lai et al., 1991; Kim et al., 2009). An exhaust gas system can have several substrates in series and parallel, each one meant to diminish a particular pollutant. Not only the computer aided optimisation of the design, but also synthesising controllers for such a complex system needs reliable, accurate and widely applicable models. The multi-zone model considered in this study requires as an input the values of local friction coefficients, which in turn depend on the geometrical features of the substrate, such as the monolith void fraction and channel cross-section shape. Those coefficients are known for circular channels, but, square, triangular and hexagonal cross-section channels are also commonly used (see Fig. 2), and the appropriate coefficients need to be determined.

The purpose of this investigation is to quantify and model the effect of the channel geometry on the pressure drop through a honeycomb type substrate using a single channel model with square, hexagonal and triangular cross-section shape. The novelty of this paper is the detailed modelling of the pressure losses through every part of the substrate, which is important for computer aided optimisation of monolith based catalytic systems. The results at a channel scale are presented in the form of the local loss coefficients for the inlet and outlet of the substrate, together with the friction factor curve along the channels for every cross-section shapes. After the model is presented, it is applied to a whole converter simulation and the results are compared to experimental data.

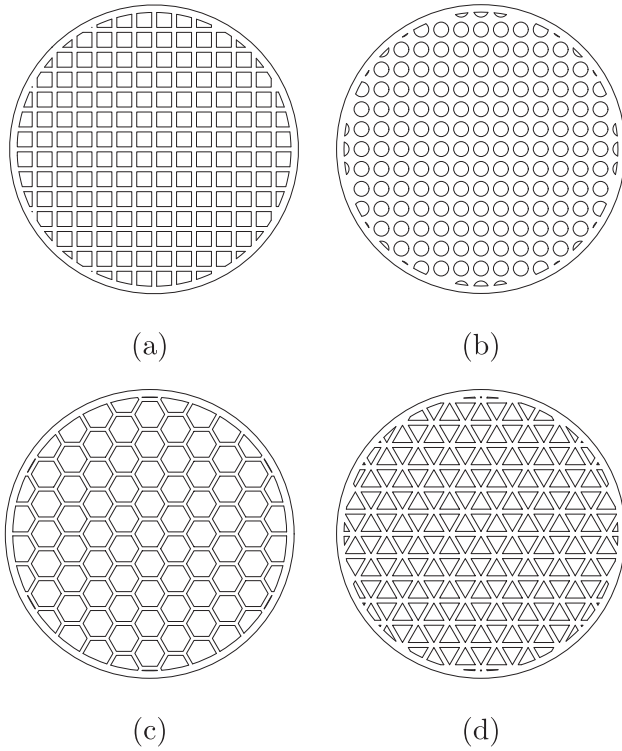


Fig. 2. Section of an extruded ceramic substrate with (a) square, (b) circular, (c) hexagonal and (d) triangular channels.

2. Computational models

This section presents the flow model, assumptions and geometrical considerations of a converter scale simulation, then for a series of channel scale numerical experiments that was used to quantify and model the pressure drop along the monolith.

2.1. Converter scale model

It is necessary to revisit converter scale flow models to build an understanding of modelling a monolith as a continuum. Fig. 3 shows a computational domain with the same specifications as the experimental set up in John Clarkson (1995). It had an inlet pipe 2000 mm long (L_i) and 54 mm in diameter ($2R_i$), followed by an inlet expansion diffuser with an angle (θ) of 35° , connecting the inlet pipe to a substrate 152 mm long (L_m) and 118 mm in diameter ($2R_m$). The substrate had square channels 1.1 mm in width with a wall thickness of 0.16 mm. Right after the substrate there was a straight Section 30 mm long open to the atmosphere (L_o). The domain is shown in Fig. 3.

2.1.1. Turbulence model

The governing equations for mass and momentum balance are (dropping the overline of the velocity average) (Ansys Fluent Ansys, 2017):

$$\frac{\partial u_i}{\partial x_i} = 0 \quad (1)$$

$$\rho \frac{\partial (u_i u_j)}{\partial x_j} = -\frac{\partial P}{\partial x_i} + \frac{\partial}{\partial x_j} \left[\mu \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} - \frac{2}{3} \frac{\partial u_l}{\partial x_l} \delta_{ij} \right) \right] - \frac{\partial \tau_{ij}}{\partial x_j} + S_{u_i} \quad (2)$$

The term τ_{ij} is the Reynolds stress tensor and depends on the flow model. In this case, the closure is provided by the Boussinesq assumption, and the SST model is used. In this model, the eddy viscosity is defined as follows (Ansys Fluent Ansys, 2017; Menter, 1994):

$$\mu_t = \frac{a_1 k}{\max(a_1 \omega, SF_2)} \quad (3)$$

The transport equations for the turbulence kinetic energy (k) and specific turbulence dissipation rate (ω) are (Ansys Fluent Ansys, 2017; Menter, 1994):

$$\rho \frac{\partial k u_i}{\partial x_i} = \frac{\partial}{\partial x_j} \left[\left(\mu + \frac{\mu_t}{\sigma_k} \right) \frac{\partial k}{\partial x_j} \right] + G_k - Y_k - S_k^{\text{sink}} + S_k^{\text{gen}} \quad (4)$$

where

$$G_k = \min(\mu_t S, 10 \rho \beta^* k \omega)$$

$$Y_k = \rho \beta k \omega$$

and

$$\rho \frac{\partial \omega u_i}{\partial x_i} = \frac{\partial}{\partial x_j} \left[\left(\mu + \frac{\mu_t}{\sigma_\omega} \right) \frac{\partial \omega}{\partial x_j} \right] + G_\omega - Y_\omega + D_\omega - S_\omega^{\text{sink}} + S_\omega^{\text{gen}} \quad (5)$$

where

$$G_\omega = \frac{\rho \alpha_\omega}{\mu_t} \mu_t S$$

$$Y_\omega = \rho \beta \omega^2$$

$$D_\omega = 2(1 - F_1) \rho 1.68 \frac{1}{\omega} \frac{\partial k}{\partial x_j} \frac{\partial \omega}{\partial x_j}$$

Further details of the blending functions and constants of the SST model can be found in Menter (1994). A realistic decay of the turbulence inside the substrate was obtained by applying the two damping terms in Eqs. (6) and (7) to the k and ω equations inside the porous medium (Cornejo et al., 2018a).

$$-S_k^{\text{sink}} = -\frac{\mu_{\text{eff}}}{\alpha_i} k \quad (6)$$

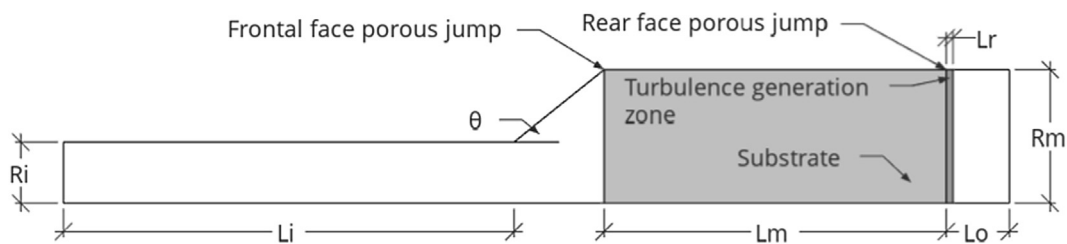


Fig. 3. Computational domain at a converter scale.

$$-S_{\omega}^{\text{sink}} = -\frac{\mu}{\alpha_i} \left(\omega_0 - C_{\omega} \frac{\mu/\rho}{\beta_{\omega} \Delta^2/4} \right) \quad (7)$$

where

$$\omega_0 = C_{\omega} \frac{\mu/\rho}{\beta_{\omega} \delta^2/4}$$

In the previous expression, C_{ω} is 6, β_{ω} is 3/4 and Δ^2 is the characteristic cell area.

It is also known that turbulence arises due to the unsteadiness produced by the flow leaving the substrate (Cornejo et al., 2018b). That phenomenon was included by adding a turbulence generation zone right after the substrate (see Fig. 3). The source terms S_k^{gen} and S_{ω}^{gen} were modelled as immersed boundary conditions, applied to the turbulence generation zone. The length scale of the turbulence l_e after the substrate was taken from discrete channel data and set as the hydraulic diameter of the channels (Cornejo et al., 2018c). Based on the same data, the dimensionless turbulence kinetic energy (k^*) was set to 0.065 (Cornejo et al., 2018c). With that, the values for k and ω in the turbulence generation zone can be estimated as:

$$k = k^* u_c^2 \quad (8)$$

$$\omega = \frac{k^{1/2}}{l_e C_{\mu}} \quad (9)$$

$\text{Re}_w = \rho u_c \sqrt{2} L_w / \mu$ is evaluated along the radius of the substrate in its rear face. The zones in which it is higher than the critical one ($\text{Re}_{w,\text{crit}} \approx 150$), known values for k and ω are imposed. Further details about the implementation of the turbulence generation after the substrate are extensively described in Cornejo et al. (2018c).

2.1.2. Permeability model

The sink term S_{u_i} in Eq. (2) applies only inside the porous zone and accounts for the extra pressure drop through the substrate. For multi-zone permeability models, the total losses of mechanical energy (ΔP_T) can be represented as the summation of the pressure drop at the frontal face of the substrate (ΔP_i), losses inside of the substrate (ΔP_s), and losses when the flow leaves the rear face of the substrate (ΔP_o). The apparent permeability inside the substrate can be accounted for by Darcy's law as (Cornejo et al., 2019b; Hayes and Fadic, 2012):

$$S_{u_i} = -\frac{\mu}{\alpha_{(x_i)}} u_i \quad (10)$$

The axial permeability is defined as the apparent permeability in the direction of the channels (i.e. flow direction) and the radial permeability corresponds to the direction perpendicular to the channel axis. For realistic results, the radial permeability should be set to be at least two orders of magnitude lower than the axial one ($\alpha_{\text{axial}} \leq 100 \alpha_{\text{radial}}$) to prevent radial flow across the continuum. The axial permeability, which is referred to as $\alpha_{(x)}$ in the rest of the paper, can be calculated as a function of the friction factor inside a monolith channel (Cornejo et al., 2019b):

$$\alpha_{(x)} = \frac{f_{(x)}}{\alpha_{fd} f_{fd}} \quad (11)$$

where

$$f_{(x)} = \frac{1}{\text{Re}} \left[\left(\frac{c_1}{\sqrt{\text{Gz}_i}} \right)^n + (f \text{Re}_{fd})^n \right]^{\frac{1}{n}} \quad (12)$$

$$\text{Gz}_i = \frac{x^+}{\text{Pr}} = \frac{x}{\text{PrReD}_H} \quad (13)$$

The subscript *fd* stands for the fully developed region, Gz_i is the inverse of the Graetz number, and c_1 and n are model constants that depend on the shape of the channels. The pressure drop due to the sudden contraction and expansion of the flow when entering and leaving the monolith can be modelled as two thin layers, or 1-D porous jumps, one at the front and other at the rear face of the monolith. The magnitude of the pressure drop through such a porous jump can be estimated as (Ansys Fluent Ansys, 2017):

$$\Delta P = \left(\frac{1}{\alpha_{\text{app}}} \mu u_i + C_u \frac{1}{2} \rho u^2 \right) \Delta m \quad (14)$$

In Eq. (14) Δm is the apparent thickness of the jump, which is used together with the apparent face permeability (α_{app}) and the inertial losses coefficient (C_u). That leads to the following expression for the losses at the front face of the substrate:

$$\Delta P_i = \left(\frac{\Delta m}{\alpha_{\text{app}}} \right) \mu u + (\Delta m C_u) \frac{1}{2} \rho u^2 \quad (15)$$

or

$$\Delta P_i = f_{1|\text{inlet}} \mu u + f_{2|\text{inlet}} \frac{1}{2} \rho u^2 \quad (16)$$

Note that f_1 accounts for the viscous losses and f_2 for the inertial ones. For the outlet of the substrate, where the inertial losses dominate, the pressure drop is:

$$\Delta P_o = f_{2|\text{outlet}} \frac{1}{2} \rho u^2 \quad (17)$$

The factors $f_{1|\text{inlet}}$, $f_{2|\text{inlet}}$ and $f_{2|\text{outlet}}$ in Eqs. (16) and (17) can be written as second order polynomials that depend on the monolith void fraction and channel geometry. The total pressure drop can be summarised by the following expression:

$$\Delta P_T = f_{1|\text{inlet}} \mu u + f_{2|\text{inlet}} \frac{1}{2} \rho u^2 + \int_{x_0}^{x_0+L_s} \left(f_{(x)} \frac{x}{D_H} \rho u^2 \right) dx + f_{2|\text{outlet}} \frac{1}{2} \rho u^2 \quad (18)$$

2.2. Channel scale model

At a channel scale, it is necessary to include the open sections before and after the substrate to investigate the pressure losses when the flow enters and leaves the substrate. A cut along the domain at a channel scale can be seen in Fig. 4.

Fig. 5 shows the frontal view of a section of a monolith for the four most common shapes. Each channel shape is accompanied by a rectangle in dashed line showing the fundamental part of each structure, which includes every feature of the geometry when symmetry is applied to the four sides.

The target void fraction for each case was obtained by manipulating the wall thickness (L_w) conveniently while keeping L constant as 1.00 mm, 1.48 mm, and 2.00 mm for the square, hexagonal and triangular channels respectively. The Re number for the channels was calculated based on the channel mean velocity and the respective hydraulic diameter, see Eq. (19). The length of the open sections before and after the substrate were enlarged systematically until no boundary effects were observed. As can be seen in Fig. N, having an open space upstream the substrate from 4 to 10 channel diameters long is sufficient for the flow rates covered in this study. Regarding the space downstream of the channel, that must be long enough to completely cover the recirculating zone after the substrate. That is from 10 to 30 channel diameters according to the literature (Cornejo et al., 2018b; Cornejo

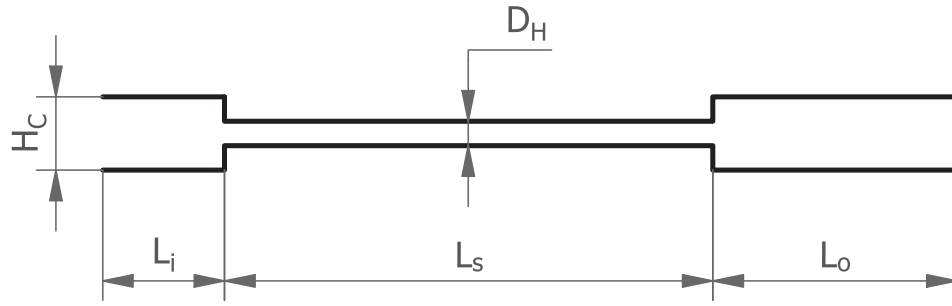


Fig. 4. Lateral view of a monolith channel with the open sections before and after it. $L_i = 20$ mm, $L_o = 60$ mm, H_c and D_H were set according to the desired channel shape and void fraction.

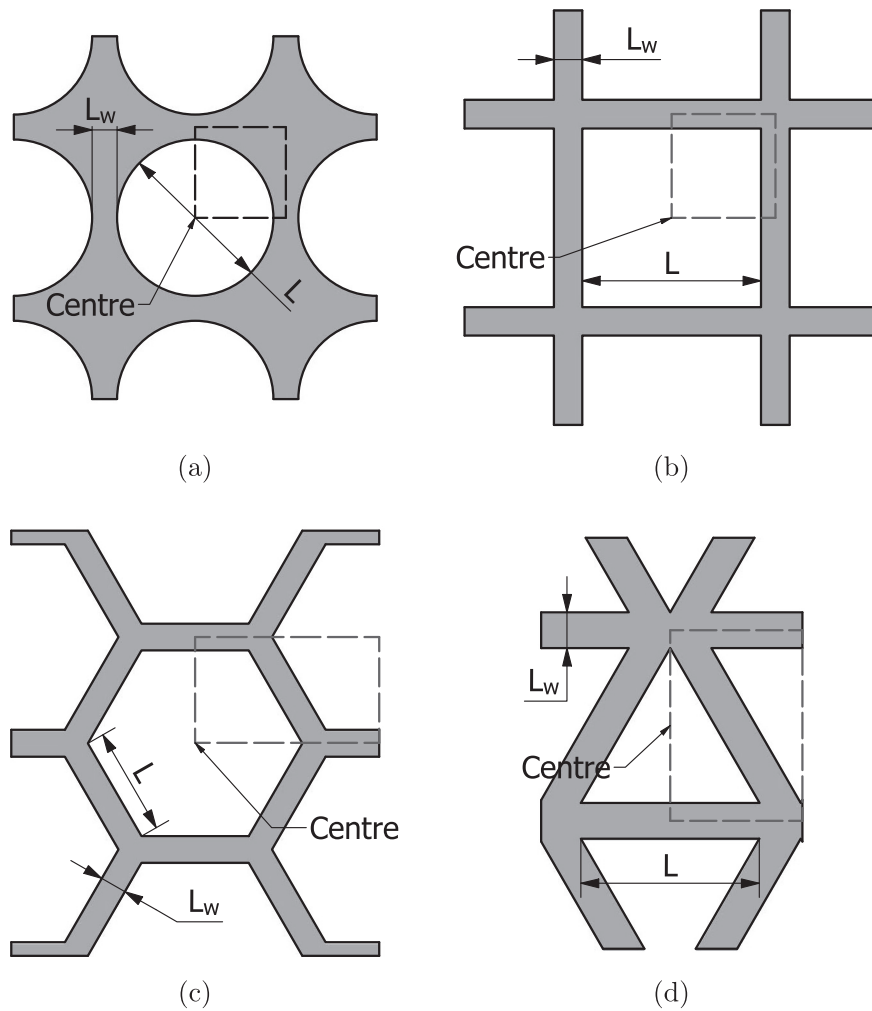


Fig. 5. Frontal view of a section of a monolith with (a) circular, (b) square, (c) hexagonal and (d) triangular channels. The centre of a channels, wall thickness and characteristic length are marked in every case.

et al., in press). In this study, we used $L_i = 20$ mm and $L_o = 60$ mm, after which length, no boundary effects were observed. The length of the substrate (or channel) was kept constant at $L_s = 40$ mm which is long enough to reach fully developed flow for all of the analysed cases.

$$Re = \frac{\rho u_c D_H}{\mu} \quad (19)$$

The working fluid was atmospheric air at 300 K. The density of the fluid was considered to be constant. This assumption was corroborated with an additional run assuming ideal gas law. The

maximum change in density observed was less than the 0.4%, and no significant changes in pressure drop were found also. However, for other scenarios, such as non-isotherm flow, using variable fluid properties is highly recommended. The flow regime was assumed to be laminar, because the channel Re is largely subcritical in most of the monolith applications. There is evidence of upstream turbulence entering the substrate in automotive applications (Ekstrom and Andersson, 2002; Cornejo et al., 2018a). There is evidence in the literature that when turbulence enters the substrate, it decays quickly inside the channels, leading to laminar flow (Cornejo et al., 2017; Cornejo et al., 2019a). Describing such

a transition accurately requires substantial computational efforts and a much more comprehensive analysis, therefore, that will be covered in a dedicated, independent paper in the future. Hence, it was desired to study the effect of the channel geometry in laminar flow first, which is valid for many applications. The resulting steady mass and momentum transport equations were:

$$\frac{\partial u_i}{\partial x_i} = 0 \quad (20)$$

$$\rho \frac{\partial(u_i u_j)}{\partial x_i} = -\frac{\partial P}{\partial x_i} + \frac{\partial}{\partial x_j} \left(\mu \frac{\partial u_i}{\partial x_j} \right) \quad (21)$$

All of the simulations were implemented in ANSYS Fluent 18.2 (ANSYS Fluent 18.2, 2017), which uses the finite volume approach. The inlet of the domain was treated as a velocity inlet, where the flow rate was manipulated according to the monolith void fraction to obtain the desired channel Re. The inner, frontal and rear walls of the section of the substrate were set as no-slip walls. The symmetry planes at the top, bottom, left and right of the domain were considered symmetric boundaries. The outlet of the domain was set as outflow. All the discretization schemes were set as QUICK and the pressure-velocity coupling was solved with the SIMPLE algorithm (Patankar, 1980). The convergence criterion was the reaching of a maximum value of scaled residuals below of 10^{-9} and having a stable total pressure drop along the entire domain. The list of all the numerical experiments covered in this study is shown as Table 1. New results are presented for hexahedral, triangular and square cross-section channels, which were also compared to those for circular channels available in the literature.

3. Results and discussion

This section describes the validation of the computational model at a channel scale, methodology of calculation, and summarises the results.

Using the outcome of the numerical experiments, the head losses at the inlet and the outlet of the substrate were calculated as the difference of the total pressure between the beginning and end of the respective section. Also, the local friction factor inside the channels was calculated based on the total head losses along them. Following the same notation as that in Fig. 4 we obtain:

$$\Delta P_i = P|_{x=0} - P|_{x=L_i} \quad (22)$$

$$\Delta P_o = P|_{x=L_i+L_s} - P|_{x=L_i+L_s+L_o} \quad (23)$$

$$f_{(x)} = \frac{\partial P / \partial x}{\frac{1}{2} \rho u_c^2} \quad (24)$$

P was calculated as the mass weighted total pressure across the corresponding transverse section of the domain, which keeps consistency with the momentum balance. Having the results of the computational experiments, they were fitted to Eqs. (12), (16) and (17) to determine the values for c_1, n, f_1 and f_2 . For f_1 and f_2 , the void fraction of the substrate, which works as the reduction and expansion rate, was computed as:

$$\phi = \frac{\text{transverse flow area inside the substrate}}{\text{transverse flow area before the substrate}} \quad (25)$$

Further details about the final expressions for pressure drop are provided in Section 3.2.3.

3.1. Grid independence and model validation

For square channels the domain was spatially discretised into a fully orthogonal hexahedral mesh. For the hexagonal and

Table 1
List of numerical experiments.

N° Run	Re	ϕ	Channel shape
1	176	0.39	Square
2	353	0.39	Square
3	529	0.39	Square
4	706	0.39	Square
5	117	0.59	Square
6	233	0.59	Square
7	350	0.59	Square
8	467	0.59	Square
9	100	0.69	Square
10	199	0.69	Square
11	299	0.69	Square
12	399	0.69	Square
13	86	0.80	Square
14	172	0.80	Square
15	258	0.80	Square
16	344	0.80	Square
17	321	0.55	Hexagon
18	642	0.55	Hexagon
19	963	0.55	Hexagon
20	1284	0.55	Hexagon
21	272	0.65	Hexagon
22	543	0.65	Hexagon
23	815	0.65	Hexagon
24	1086	0.65	Hexagon
25	232	0.76	Hexagon
26	465	0.76	Hexagon
27	697	0.76	Hexagon
28	929	0.76	Hexagon
29	210	0.84	Hexagon
30	420	0.84	Hexagon
31	631	0.84	Hexagon
32	841	0.84	Hexagon
33	132	0.60	Triangle
34	265	0.60	Triangle
35	397	0.60	Triangle
36	530	0.60	Triangle
37	120	0.66	Triangle
38	241	0.66	Triangle
39	361	0.66	Triangle
40	482	0.66	Triangle
41	109	0.73	Triangle
42	218	0.73	Triangle
43	327	0.73	Triangle
44	435	0.73	Triangle
45	95	0.84	Triangle
46	189	0.84	Triangle
47	284	0.84	Triangle
48	378	0.84	Triangle

triangular cross-section channels, polyhedral computational cells were used together with an inflation layer at the walls. Fig. 6 shows a hexahedral and a polyhedral mesh for the square and hexagonal channels respectively. Since the area of the inlet of the domain was manipulated to obtain the desired set of void fractions, the total volume of the domain slightly changed between cases. However, every mesh had of the order of four million computational cells.

The grid independence of the solutions was investigated by refining the mesh systematically and monitoring the total pressure drop along the domain. In a square channel geometry identical results were obtained for two meshes of 4.2 and 11.9 millions of cells each. The same test was performed for hexagonal and triangular channels leading to equivalent results. In addition, the quality of the solution was analysed in two ways simultaneously, monitoring the hydraulic entrance length (L_H) and the friction factor ($f_{Re_{fd}}$) at the fully developed region.

According to the results in Fig. 7, the computational model was able to converge exactly to the correct friction factor in the fully developed region for all of the channel Re and channel shapes covered in this study. That constitutes an important validation,

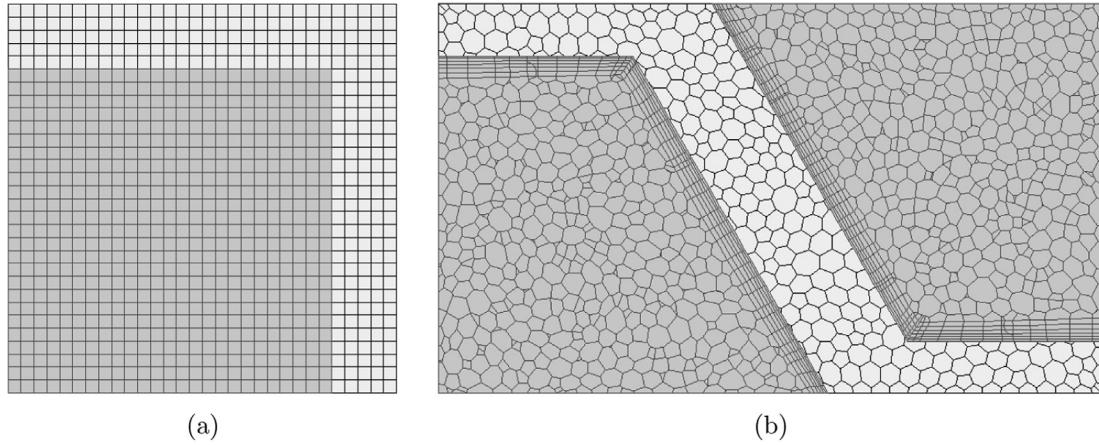


Fig. 6. Mesh for a section of a (a) square channel and (b) hexahedral channel. The grey cells represent the mesh inside the channels.

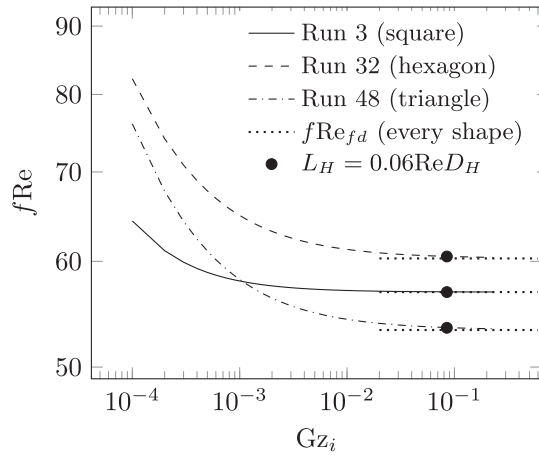


Fig. 7. Friction factor for three channel cross-section shapes. The expression for L_H was taken from Bergman et al. (2011) and the fRe_{fd} for square and triangular pipes from White (2009).

because the pressure drop is the main focus of this study and it is a function of the friction factor. In addition, for the hydraulic entrance length, the numerical model showed consistency with the expected, according to literature for the three channel shapes. For honeycomb monoliths, the reduction of the flow area entering the substrate produces a non-flat velocity profile at the inlet of the channels (Cornejo et al., 2019b; Benjamin et al., 2001). That has an effect on the extension of the developing region, however, expressions assuming a flat inlet velocity profile still provide a useful approximation to investigate consistency.

3.2. Effect of channel geometry on the pressure drop

This section summarises the pressure losses for flow through a substrates. Fig. 8 provides a first sight to the flow entering and leaving a channel. In this section, the pressure drop in those zones, and also inside the channels, is analysed in detail for several flow rates and channel cross-section shapes.

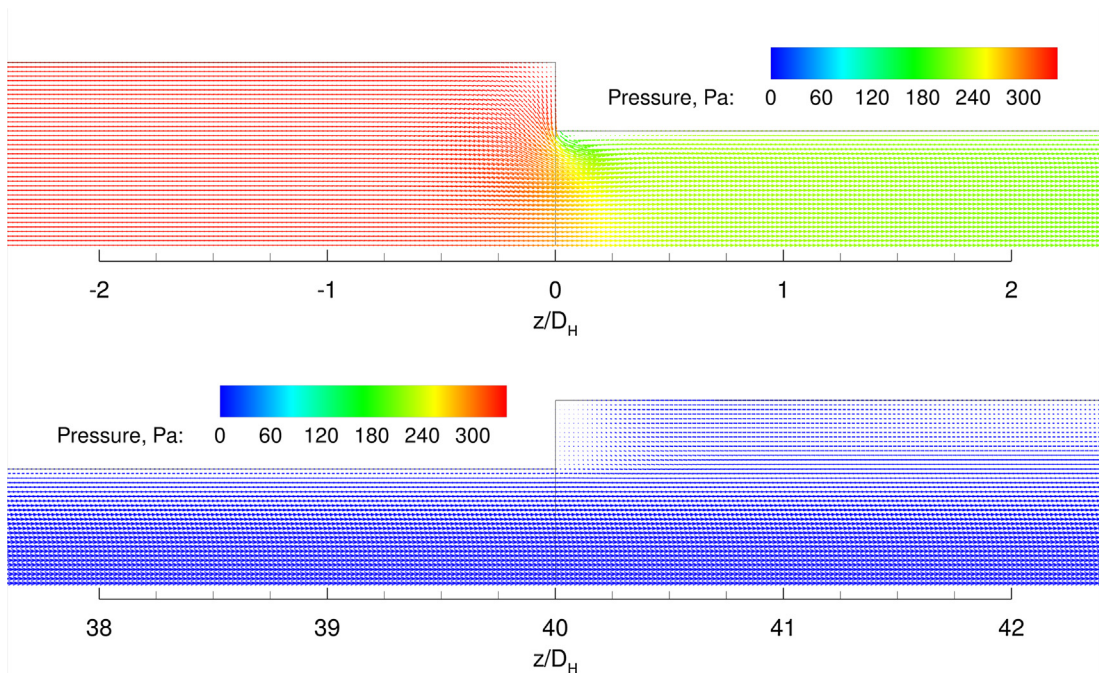


Fig. 8. Lateral view of velocity vectors on the symmetry plane of a square channel colored by static pressure. Inlet and outlet zones. $Re = 399$ and $\phi = 0.69$ (Run 12).

3.2.1. Losses when entering and leaving the substrate

Fig. 9 shows the pressure losses for flow entering and leaving the substrate for several void fractions, channel shapes and channel Re. Consistent with that reported for circular pipes, the head losses have a quadratic dependence on the channel velocity. Also, the magnitude of the losses is inversely proportional to the substrate void fraction (or directly proportional to the reduction rate). When entering the substrate, the magnitude of the losses for hexagonal and triangular channels are comparable and they are significantly lower than those for square channels. Also, the total pressure drop when leaving the substrate was always higher than that when entering the substrate, for a given void fraction and channel velocity.

Each data set shown in Fig. 9 is accompanied by an individual fitting to Eqs. (16) and (17) correspondingly. That allowed us to calculate f_1 and f_2 as a function of the monolith void fraction. The resulting expressions are shown in Table 2 and make it possible to use Eqs. (16) and (17) to describe every curve in Fig. 9 accurately. The sensitivity of f_1 and f_2 to the channel shape and monolith void fraction can be seen in Fig. 10. All of the individual fits showed an R^2 above 0.97 and are useful to interpolate f_1 and f_2 for ϕ values between 0.5 and 0.9.

3.2.2. Losses inside the substrate

Regarding the pressure drop inside the substrate, the curve f -Re- Gz_i was calculated for every case in Table 1. The resulting data

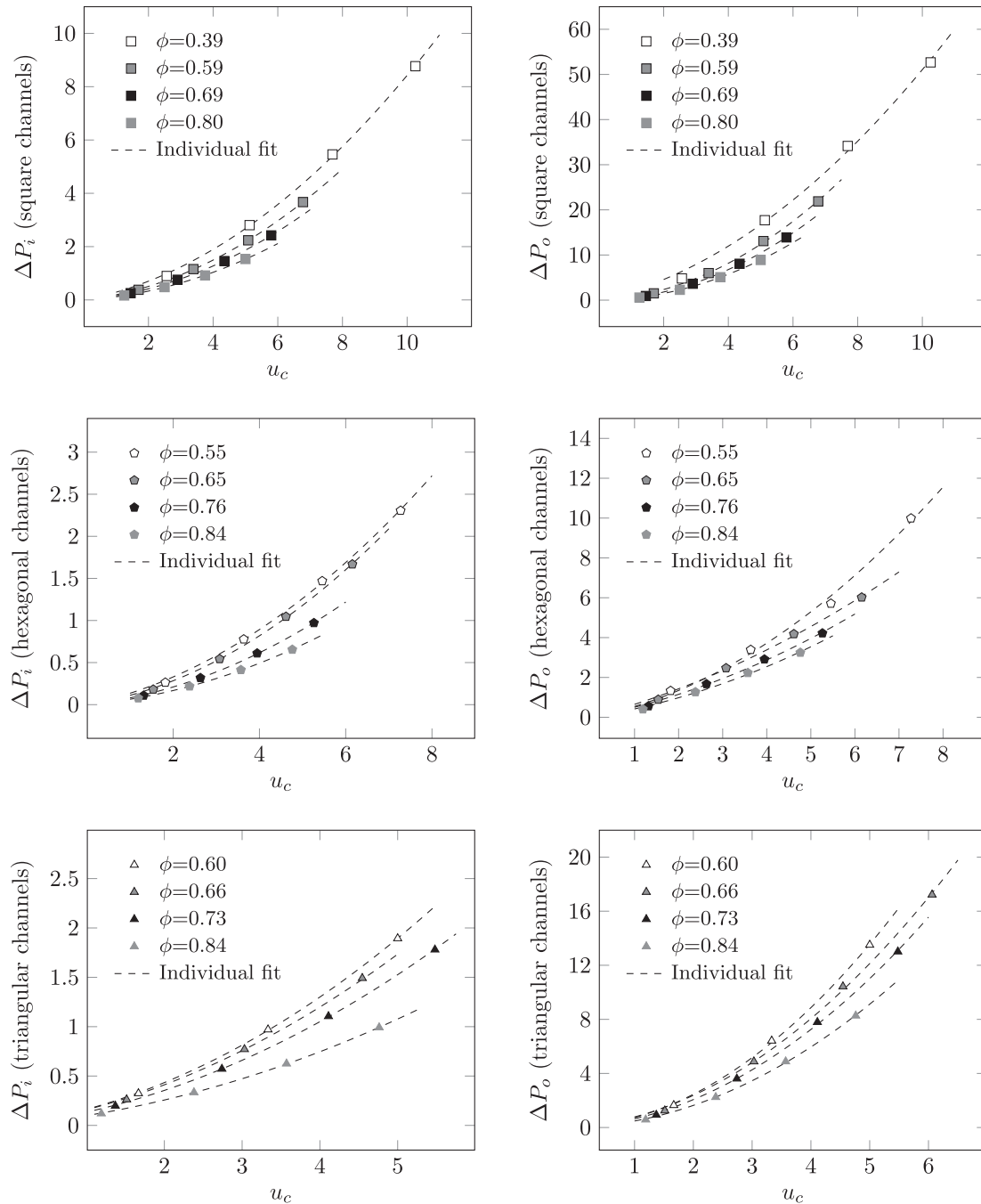


Fig. 9. Head losses for flow entering (left) and leaving (right) the substrate for several void fractions and channel velocities. ΔP is in Pa and u_c in m/s.

Table 2

Multi-zone permeability parameters for several channel shapes.

	Circular (Cornejo et al., 2019b)	Square
$f_1 _{inlet}$	$-6678.9\phi^2 - 8294.9\phi + 16956$	$25868\phi^2 - 52117\phi + 29224$
$f_2 _{inlet}$	$-0.127\phi^2 + 0.0590\phi + 0.0816$	$-0.2279\phi^2 + 0.2119\phi + 0.0523$
$f_2 _{outlet}$	$0.9782\phi^2 - 2.5968\phi + 1.9567$	$-0.3137\phi^2 - 0.6540\phi + 1.3131$
c_1	2.5855	2.3644
n	2.1154	2.1156
fRe_{fd}	64.0	56.9
	Hexagonal	Triangular
$f_1 _{inlet}$	$17395\phi^2 - 36145\phi + 20594$	$-17021\phi^2 + 15493\phi + 2950.3$
$f_2 _{inlet}$	$-0.2055\phi^2 + 0.2320\phi - 0.0178$	$-0.1918\phi^2 + 0.1084\phi + 0.0918$
$f_2 _{outlet}$	$-0.5490\phi^2 + 0.4617\phi + 0.3364$	$0.8620\phi - 2.4613\phi + 2.0943$
c_1	1.9048	2.4701
n	1.7270	1.9756
fRe_{fd}	60.3	53.3

are shown in Fig. 11. As observed previously for circular channels (Cornejo et al., 2019b), the effect of the void fraction and channel Re on the fRe curve is present but moderated in magnitude. Due to the reduction of the flow area, the velocity profile at the inlet of the substrate changes slightly and the size of the stagnation zone right after the inlet of the channels remains approximately constant. As expected, the fRe curve for every shape tends asymptotically to the corresponding value at the fully developed zone. However, when the curves are scaled by their respective asymptotic value, all of them follow a similar trajectory. That is shown in Fig. 11.

3.2.3. Summary of the pressure drop model coefficients

Table 2 summarises the expressions and parameters to compute the pressure drop of flow entering, within and leaving the substrate for the three channel shapes investigated in this study. Data for circular channels previously reported in the literature were also included.

4. Implementation at a converter scale

Having defined the model parameters for several channel shapes, this section shows the implementation of the permeability model at a converter scale and compares its prediction to experimental data available in the literature for a substrate with square channels.

4.1. Flow model

The domain and turbulence model at a converter scale, which included an inlet pipe, a diffuser and a monolith, are described in Section 2.1. For the apparent permeability of the substrate, two models for the source term S_{u_i} in the momentum balance in Eq. (2) were considered. The first one assumed Poiseuille flow, where the apparent permeability of the substrate was assumed constant and it was estimated as follows:

$$\alpha_{axial} = \frac{\phi D_H^2}{28.4} \quad (26)$$

$$\alpha_{radial} = \frac{\alpha_{axial}}{100} \quad (27)$$

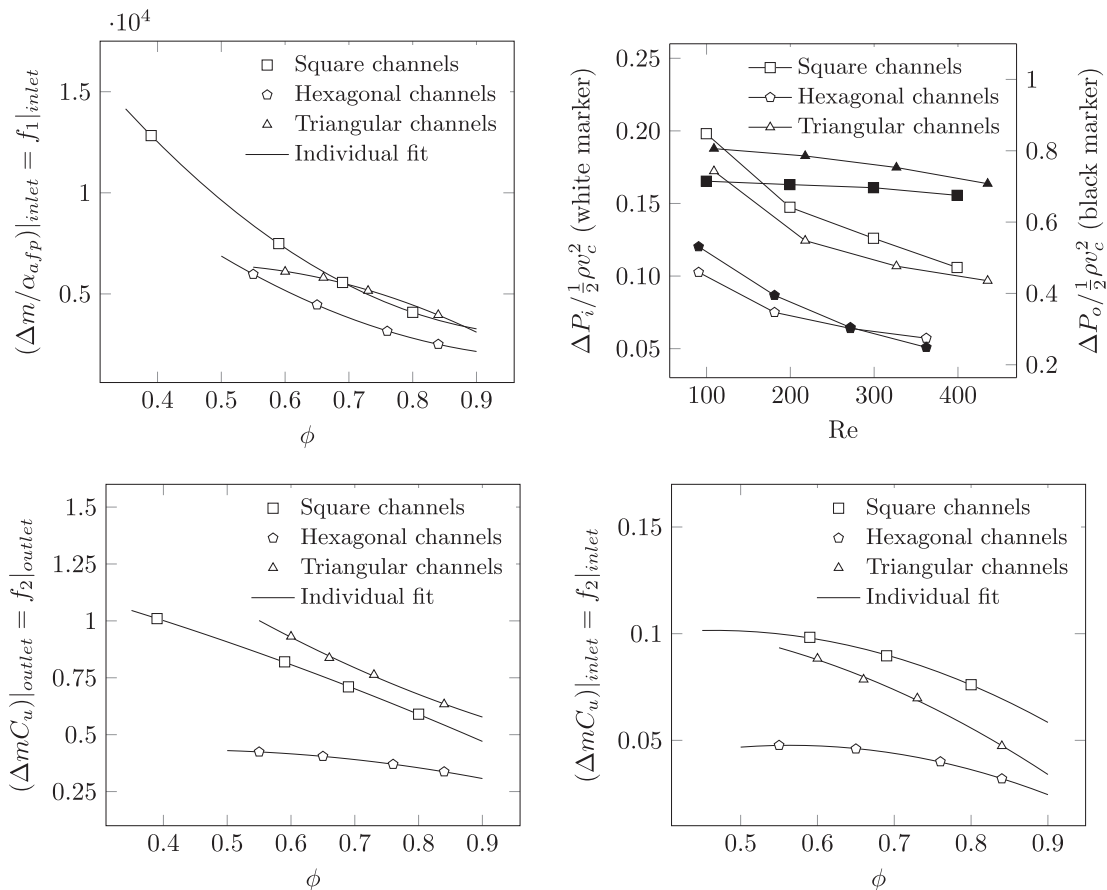


Fig. 10. Parameters for the local friction coefficient and apparent face permeability for several channel shapes and void fractions. f_1 is in 1/m.

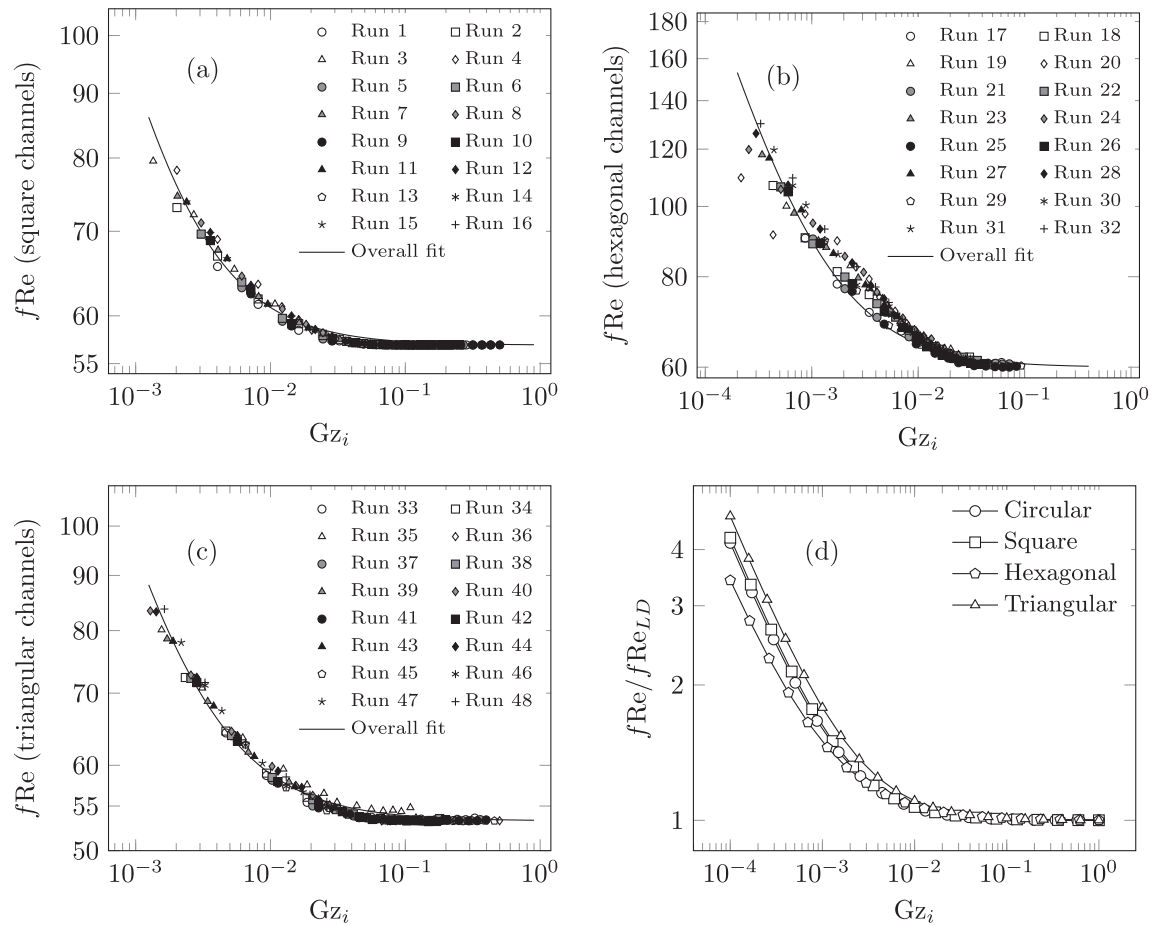


Fig. 11. Friction factor for (a) square channels, (b) hexagonal channels and (c) triangular channels. (d) shows the scaled friction factor for four channel shapes simultaneously. The run number refer to those in Table 1.

The second option was the multi-zone permeability model, with the parameters presented in this work. It accounts for the pressure drop at the inlet, outlet and along the substrate separately. One porous jump was set at the frontal face of the substrate to account for ΔP_i and another at the rear face to account for ΔP_o . Both porous jumps are 2D surfaces, where the pressure drop occurs immediately once the flow passes through them. The equations for the two jumps are Eq. (16) and (17) respectively. The values for f_1 and f_2 for square channels are given in Table 2. One of the main drawbacks of the continuum approach is that, in contrast to a real monolith substrate, there are no channel walls or changes in the flow area, hence, phenomena such as the flow acting as a series of jets leaving the substrate are not represented correctly. Porous jumps are a strategy that allows us to account, at least, for the pressure drop when entering and leaving the substrate. At the frontal face of the substrate, energy losses happen in a thin zone, hence, a porous jump is reasonable. The exit of the substrate requires more attention because the recirculating flow. Having a detailed model of a progressive pressure drop of the flow moving downstream would be ideal in scenarios in which there is a second element in series, close enough to modify the recirculating zone and pressure drop of the first substrate. A model like that, able to describe jet flow leaving the substrate, requires extensive further research and is beyond the scope of this paper. The permeability inside the substrate was calculated using Eq. (11) with the parameters for square channels given in Table 2.

4.2. Solver settings and grid analysis

As previously, the simulations were implemented in ANSYS Fluent 18.2 (ANSYS Fluent 18.2, 2017). The inlet was a velocity inlet,

with a velocity magnitude according to the desired inlet Re. The inlet turbulence intensity was set as 15% and the turbulence scale as $0.4D_H$, which are recommended values for fully developed turbulent flow in pipes (Ansys Fluent Ansys, 2017). The outlet was treated as a pressure outlet. Advantage was taken from the axial symmetry of the geometry, hence, the centre line of the domain was set as symmetry axis. The outside boundary of the domain was considered a no-slip wall.

The operating fluid was incompressible atmospheric air at 300 K, the same as in John Clarkson (1995). The schemes for momentum, k and ω were set as QUICK, and the pressure velocity coupling was solved with the SIMPLE algorithm. The convergence criterion was having scaled residuals with a value below 10^{-6} and a stable total pressure drop and volume average velocity magnitude. The domain was discretized in a fully quadrilateral mesh of 240 000 cells. Another mesh with 480 000 cells was also tested, producing a total pressure drop, volume average velocity magnitude, and velocity profile after the substrate that differ by less than 0.25% from those obtained with the previous mesh.

4.3. Results at a converter scale

Fig. 12 shows the velocity profile after the substrate when assuming Poiseuille flow and when using the permeability model presented in this work compared to the experimental presented in John Clarkson (1995).

Two inlet Re were tested. It can be seen that the new model leads to much better agreement in the velocity profile for both inlet Re. Compared to the case when assuming Poiseuille flow, this

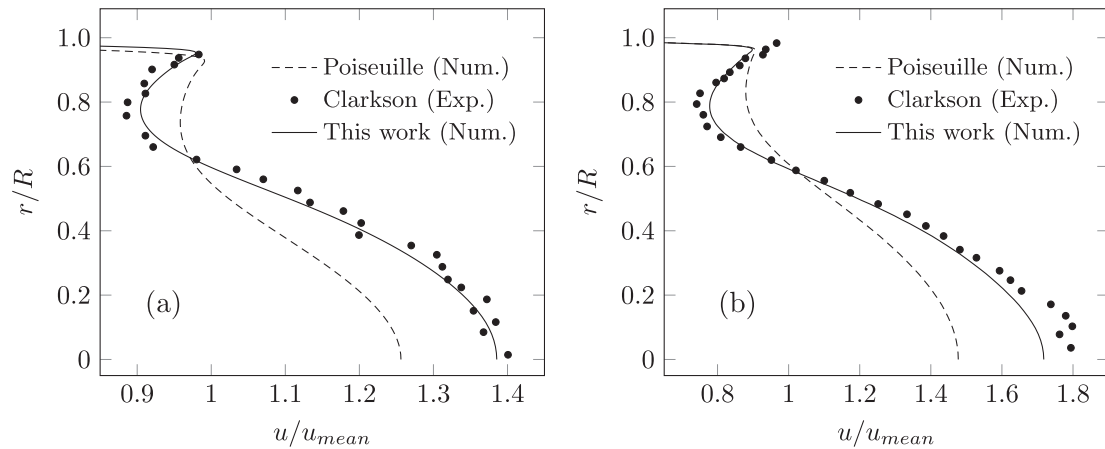


Fig. 12. Comparison of the predicted and experimental velocity profile after the substrate at (a) inlet Re 30,000 and (b) inlet Re 60,000.

work improves the prediction of the main peak velocity and the secondary peak close to $r/R = 1$. Both regions of the velocity profile are a common issue when modelling a converter, because current models tend to predict velocity profiles flatter than those observed experimentally. We emphasize that the improvement in the prediction of the velocity profile achieved in this paper comes from a channel scale model. No parameters were fit using converter scale data, hence, the predicted and experimental velocity profiles are two completely independent data sets; the model is purely predictive.

It is well established in the literature that a higher pressure drop leads to a comparatively flatter velocity profile. Following that logic, and taking a closer look at the experimental data, it can be seen that the measured velocity profile is more mal-distributed than that from when assuming Poiseuille flow. Consequently, one can conclude that the experimental pressure drop must be smaller than the one when assuming Poiseuille flow. It might be counter-intuitive that after adding extra flow resistance, such as the one from the entering and leaving effects, the flow becomes more mal-distributed. However, contrary to the case for the flow through a unique channel, in a whole monolith, there is an additional degree of freedom, which is precisely the flow distribution. For the example in Fig. 12, despite of the significant differences in flow distribution, the total pressure using a variable and a constant pressure drop were comparable (see Table 3). This point requires especial attention, together with an extensive and detailed experimental validation, which will be addressed in a further paper.

It can also be seen that at an inlet Re of 60 000 there is still a small underestimation of the peak velocity. That might be explained by other phenomena not considered in this work, such as the effect of the upstream turbulence on the apparent friction factor inside the channels and the flow entering in an oblique angle to the substrate. Back pressure and flow distribution are interrelated, hence an improvement in the flow distribution due to a better permeability model is expected.

5. Conclusions

The effect of the channel geometry on the pressure drop through a monolith honeycomb was investigated using an extensive set of computational experiments. Local friction coefficients for every investigated channel cross-section shape and monolith void fraction were presented in the form of an apparent permeability model for full monolith scale simulations and also as a semi-empirical lumped model for total pressure drop. A consistent first principles based methodology was proposed for developing pressure drop equations in monolith systems.

Overall, the findings were consistent with those observed previously for circular channels; the monolith void fraction and approach velocity affect the shape of the velocity profile at the inlet of the channels. However, the change in the inlet velocity profile was observed to be relatively small, and had only a minor effect on the fRe vs. Gz_i curves. Therefore, for each channel shape, a single fRe vs. Gz_i curve was sufficient to describe the friction factor with an insignificant loss in accuracy. It should be emphasized, however, that use a flat inlet velocity profile leads to significant error. Additionally, the fRe curves for the different channels shapes follow a comparable trajectory when they are scaled by their respective asymptotic value.

The new model gives much better agreement with the experimental full-scale velocity profiles than those obtained when assuming Poiseuille flow and a constant apparent permeability. It improves significantly the prediction of the velocity profile across the substrate, especially the main velocity peak at the centre of the substrate and the secondary one, close to the outer boundary. Although the contribution of the so-called minor losses to the total pressure drop is low in percentage, it has been proven that in complex systems, such as a honeycomb substrate, accounting for them improves the accuracy of the model significantly. The channel shapes investigated showed a similar dependence on the substrate void fraction and approach velocity. However, the magnitude of the pressure drop and local friction coefficients were specific for every shape. Since the flow distribution impacts strongly other important variables in catalytic reactors, such as the temperature distribution, conversion efficiency and warm-up time, other channel shapes, for example, those of washcoated channels should be investigated using the same approach. In critical applications, a shape specific correlation should be developed using the reported methodology for the best prediction of the pressure drop and resultant velocity profile.

Table 3
Pressure drop through the substrate.

Inlet Re	Permeability Model	Pressure drop, Pa
30 000	Poiseuille flow	157.2
30 000	This work	157.4
60 000	Poiseuille flow	344.1
60 000	This work	343.1

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

I. Cornejo acknowledges the receipt of a Becas-Chile (CONICYT) scholarship. Other funding was provided by the Natural Sciences and Engineering Research Council of Canada. This research was enabled in part by support provided by Compute Ontario (computeontario.ca) and Compute Canada (www.computeCanada.ca).

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